	Azuresky	
	Management system requirements	Document no: ASDP001
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

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Azuresky (Pty) Ltd is entering the coal laboratory testing market with the aim of expanding the range of available services and providing a competitive, high-quality alternative within the industry. The company is committed to delivering reliable, accurate, and technically sound results, while continuously improving its service offering to meet customer needs and maintain a strong position in the market.

Services Offered

Azuresky (Pty) Ltd provides the following services:

- Proficiency Testing (PT) Schemes
  Design and operation of proficiency testing schemes conducted in multiple rounds for coal testing, measurement, calibration, and inspection activities, in accordance with applicable standards.
- Certified Reference Material (CRM) Production
  Production of certified reference materials that are prepared, tested, and verified to meet required homogeneity and stability criteria, ensuring fitness for purpose.
- Coal Testing Services
  Laboratory testing of coal samples using recognised methods, including:
  - ISO Ash [1171] (or Quick Ash)
  - Inherent Moisture [SANS 5925]
  - ISO Volatile Matter [562]
  - ISO Calorific Value [1928]
  - ISO Moisture [589]
  - Particle Size Distribution (PSD) [1953]
  - Float and Sink Analysis [7936]

The purpose of management system development

The purpose of the management system is to identify, design, and plan all activities that directly affect the validity of **testing, proficiency testing (PT) schemes, and certified reference material (CRM) production**, and to ensure that these activities are carried out in a consistent and controlled manner in accordance with **ISO/IEC 17025:2017**, **ISO 17043:2023**, and **ISO 17034:2016**.

The laboratory shall ensure that all relevant processes are defined/documented, implemented, and maintained to support the accuracy, reliability, and traceability of results. All activities shall be performed using approved methods, competent personnel, and suitably calibrated and verified equipment.

Any significant changes to the initial or existing design, methods, equipment, materials, or processes that may impact the validity of testing, PT schemes, or CRM production shall be identified, reviewed, and authorised prior to implementation (where applicable and/or necessary). A risk-based approach shall be applied (where necessary) to assess potential impacts, and appropriate actions shall be taken to eliminate or minimise such risks. Records of these evaluations and actions shall be retained.

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1. General requirements

1.1 Impartiality

ISO 17043:2023 clause 4.1, ISO 17034:2016 clause 4.2 and ISO 17025:2017 clause 4.1 require that the activities in the organisation are undertaken impartially.
The organisation has structured to manage impartiality in developing three documents:

- Risk assessment to impartiality
- Impartiality policy
- Declaration of interest

The **Risk Assessment to Impartiality (Document ASRI-4.2-001)** was developed and implemented to identify risks associated with laboratory activities, including coal testing, proficiency testing schemes, and reference material production.
The process involves:

- Identifying risks by considering potential future events that may negatively impact individuals, assets, customers, or the environment.
- Evaluating each risk in terms of likelihood and consequence.
- Defining mitigation actions to reduce or eliminate adverse effects.

Risk Classification & Response

- When a risk materializes, it is treated as a **non-conformance** and addressed in accordance with **Procedure 7.17 (Non-Conforming Work)**.
- Management convenes to evaluate the extent of the risk's impact on laboratory activities and/or customers. Risks identified through ASRI-4.2-001 are classified as **major**.
- Potential risks projected into future events are classified as **minor**, as they represent anticipated scenarios rather than current non-conformances.
- Monitoring and follow-up periods are aligned with the requirements of **Procedure 7.17**, ensuring consistency in corrective action and review.

This structured approach ensures impartiality is safeguarded, risks are systematically managed, and corrective actions are traceable and effective.

Impartiality policy document no. ASMF 4.2 was developed to formally document and outline the standards and expectations within the organization, providing clear guidance for employee behavior and decision-making. It helps eliminate micromanagement by setting clear directions, improving efficiency, and reducing stress in human resources management. Well-defined policies promote professionalism both internally, fostering a respectful work environment, and externally, enhancing the organization's reputation and business relationships.

Declaration of interest document no. ASMF 4.2.1 was developed to promoting transparency and good governance in the organisation. Again, to ensure that decisions are made in the best interests of the company and that any potential conflicts of interest are identified and managed appropriately. All team members are obligated to complete and sign the document.

- All personnel shall complete the declaration of interest document ASMF 4.2.1 during the onboarding process (within a week after being hired).

1.2 Confidentiality

ISO 17043:2023 clause 4.2, ISO 17034:2016 clause 4.3 and ISO 17025:2017 clause 4.2 require that the organisation shall be responsible, through legally enforceable agreements for the management of all information obtained or created during the testing services, the running of proficiency testing scheme and reference material production.
The organisation has structured to manage confidentiality in developing three documents:

- Quality Policy Statement document no. ASMF 8.2
- Management of confidential information document no. ASMF 4.3
- Non-disclosure agreement document no. ASMF 4.3.1 was designed to be signed by all personnel involved in testing, proficiency testing scheme and reference material production.
- Onboarding of new personnel shall include the completion of non-disclosure agreements document no. ASMF 4.3. within a week of onboarding.

1.3 Structural requirements

Azuresky Propriety Limited is registered with Companies and Intellectual Property Commission (CIPC), the registration number is 2024/223777/07. The business is located in address Unit 3, 17 Makriel Street, Wadeville, GERMISTON 1428.

Azuresky PT scheme sources coal, prepares and test as per the guidelines of ISO 17025:2017, then package and ships it for activities of the PT scheme as per the guidelines of ISO 17043:2023 and produces the reference material as guided by ISO 17034:2016. Therefore, the business operates as per the guideline of these three ISO standards.

There are three positions defined in the management structure/organogram that are responsible and accountable for the running of the PT scheme and reference material

production, as well as testing. These positions of a Laboratory Supervisor, Laboratory Manager and Managing Director.

Azuresky laboratory facility conforms and participates in Coal concept PT scheme, participation is on the following testing methods:

- Inherent Moisture
- Volatile matter
- Quick/ISO Ash
- Calorific Value a
- Total Moisture

The organisation does have other personnel as presented on the organogram who carry out their duties which include:

- In-house maintenance e.g., maintenance of CV bombs, crusher and pulveriser efficiencies etc. These are Laboratory Analysts and Laboratory operators.
- Development and improvement of management system; Laboratory Manager, Operations Manager, and Laboratory Supervisor.
- Except through internal audit process, all personnel are responsible to identify the deviations from the management system or from the procedures and initiate actions to prevent or minimise such deviations (they are required to report to management on the performance of the management system).
- The task of performing internal audit is allocated to be performed by an external auditor or MD of the company or any personnel deemed competent by management.

During the development and implementation of the management system, it was essential to ensure effective communication- maintaining attendance registers for meetings was required as evidence of this engagement. This guaranteed the participation of all personnel in understanding parts of the system relevant to them.
All personnel involved in testing activities, proficiency testing (PT) and reference material (RM) production are granted access to relevant sections of the management system documentation and related information, according to their responsibilities. Documentation is stored on the QMTEK server in designated folders. Access permissions are applied to ensure confidentiality, with restricted folders (e.g., HR records).
This approach ensures that all staff remain aware of processes contributing to testing, PT and RM activities, while safeguarding sensitive information through controlled access.

2. Resource Requirements

This is as per ISO 17043, ISO 17034 and ISO 17025 clause 6

2.1 Personnel

The business aimed to start productively and efficiently; it was intentional to search and build a management team that is skilled in coal analysis testing methods (personnel who worked at the coal laboratory and testing coal) and who have obtained qualification in Analytical Chemistry or Chemical Engineering or Geology.

Recruiting the right people not only has a positive impact on productivity and performance but also helps reduce costs associated with turnover and poor performance. The calibre of such personnel train new employees and manage potential disruptions to operations. Additionally, high-performing employees deliver a strong return on investment (ROI)

The composition of team was characterised as follows in terms of education/qualification:

- Laboratory Manager /Laboratory Supervisor: Qualification in Analytical Chemistry or Chemical Engineering or Geology (+ relevant experience)
- Laboratory Analyst: Undergraduate qualification in Analytical Chemistry or Chemical Engineering or Geology
- Laboratory Operator: Grade 12 qualification.

The organogram of the organisation is diagrammatically presented in Document no. ASMF 5.1. All functions in the organogram have job descriptions established and implemented. It is a requirement that personnel occupying the position shall sign the document and contract of employment. This is communication to personnel of their duties, responsibilities and authorities.

- **Managing Director**: Job description title Managing Director document no. ASHR 6.2.2 E lists the responsibilities associated with the post.
- **Laboratory Manager**: Job description title Laboratory Manager document no. ASHR 6.2.2 D lists the responsibilities associated with the post.
- **Laboratory Supervisor**: Job description title Laboratory Supervisor document no. ASHR 6.2.2 C lists the responsibilities associated with the post.
- **Laboratory Analyst**: Job description title Laboratory Analyst document no. ASHR 6.2.2 B lists the responsibilities associated with the post.
- **Laboratory Operator**: Job description title Laboratory Operator document no. ASHR 6.2.2 A lists the responsibilities associated with the post.

The action of selection of personnel in the organisation shall be addressed in accordance with the requirements of document ASMF6.1 TS. This procedure also addresses the requirements of obtaining operators/analyst through in-service programme.

The organisation has developed and retained records of monitoring of competence of personnel. All competence and authorizations must be reviewed annually or when there are any role changes. Records must be kept for 2 accreditation cycles. Procedure for the declaration and verification of personnel competence ASMF6.2 has also stated period/condition for the review of competence period for the retention of records. Personnel declaration form ASMF6.1.1 shall be utilized to declare personnel.

NB: Supervisor/ Laboratory Manager shall provide training to personnel in the organisation. If necessary, consultant may be requested to provide training, and this will be conducted in compliance to the conditions stated for externally provided services. When the company has been well established and the other personnel are deemed to be competent, they can take over the responsibilities of training others.

The Laboratory Manager and Laboratory Supervisor are responsible for all **testing activities**, including the preparation and evaluation of PT items and the production of reference materials. Their duties encompass:

- Assessing data to determine stability, homogeneity, assigned values, and associated uncertainties of PT items and reference materials.
- Evaluating the performance of PT participants.
- Providing opinions, interpretations, and technical advice to participants (this only applies to the PT function).

- Compiling reports and certificates related to testing, PT activities, and reference materials.

The **Operations Manager** is responsible for the **review and authorization** of all compiled information, ensuring that PT reports, testing reports, and reference material certificates are formally validated before release.

This structured allocation of responsibilities ensures accountability, technical rigor, and compliance with ISO/IEC 17043, ISO/IEC 17025, and ISO 17034 requirements, while maintaining transparency across testing, PT, and reference material activities.

2.2 Facilities and environmental conditions

The scope of operations includes **sample preparation of coal samples, testing and analysis, packaging, and storage**. The available and allocated operational space has been planned, partitioned, and demarcated to include:

- A sample preparation area
- A room housing furnace
- A room housing the CV machine and inherent moisture oven
- A storage room for finished products (PT samples and certified reference materials)

The **Laboratory Manager and Laboratory Supervisor** are responsible for all testing activities, including the preparation and evaluation of PT items and the production of reference materials. Their duties cover:

- Assessing data to determine stability, homogeneity, assigned values, and associated uncertainties of PT items and reference materials.
- Evaluating PT participant performance.
- Compiling reports and certificates for testing, PT activities, and reference materials.

The **Operations Manager** reviews and authorizes all compiled reports and certificates (for PT and CRMs only), ensuring formal validation before release.

Factors affecting the validity of results—such as **temperature control and other environmental conditions**—have been defined and documented in the procedure *Environmental Control* (Registration No. ASMF 7.2.3). This procedure establishes monitoring, control, and recording requirements for environmental conditions.

Minimum signage for safety requirements, particularly regarding personal protective equipment (PPE), has been defined and affixed to the walls in accordance with the scope of operations.

2.3 Externally provided products and services

Azuresky (Pty) Ltd seeks to enhance the production of its activities and obtain externally provided products and services. As guided from ISO 17025:2017 clause 6.6 and 7.1, ISO 17034 clause 6.2 and ISO 17043 clause 6.4, it is the company's intention to incorporate externally provided products and services into laboratory's activities. They are intended to support the operation of the laboratory.

They include but not limited to the following products and services:

- Measurement standards e.g., Benzoic acid, etc
- Service and repair of equipment
- Mass and Temperature calibration of equipment
- Laboratory information management system
- Proficiency Testing Scheme(s)
- Consumable materials, equipment, supply of equipment components
- Assessment and auditing services

- Manpower: a service level agreement has been entered into between Azuresky (Pty) Ltd and Hlogirati (Pty) Ltd for the provision of 4 (four) undergraduate students.

NB: The condition for acquiring the service of mass and temperature calibration of equipment, and service and repair of equipment, competent (not necessarily accredited) service providers are authorized to be used to provide the service.

The checklist (covered in equipment procedure 6.4) titled "Provision of equipment, services and supplies" number ASMF6.6.3 was developed and shall be used to actively select, monitor, measure, and manage the performance of a company's supplier to ensure they meet the requirements and acceptance criteria as set out in their respective checklist. When a supplier has been approved according to the criteria & requirements, the agreement is valid for a year. When it elapses, management shall evaluate the service received, and if it meets the stipulated requirements & no issues have arisen during the year, the agreement rolls over to the following year. Upon re-evaluation, a section will be added to the checklist as proof of re-evaluation.

Disclaimer: Management reserves the right to evaluate a service provider at any time during the stipulated period of validity if critical issues arise with the agreement in place or a new and better service provider becomes available. The new service provider shall undergo the same process before approval.

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3. Process requirements

3.1 Establishing, contracting and communicating the PT scheme objectives

The procedure to handle contractual matters, requests and tenders was developed BUT has not been implemented because the lab does not currently have any clients to the testing part. The document number is ASMF 4.7.1. It seeks to address all requirements of ISO 17043:2023 clause 7.1.1, ISO 17034:2016 clause 4.1.1, ISO 17025 clause 7.1.

There are no parts of the running of the PT scheme and reference material production undertaken by subcontractors. All parts are performed at the company.
AS026TF Outsourced test form was developed and implemented to be used for the purpose of subcontracting tests.
ASMF 6.6.3 Externally provided products and services procedure was developed and implemented for use of acquiring external products and services.

3.2 PT scheme communication

Azuresky PT scheme roll out plan document number ASMF 7.2.1 PT Plan (PRODUCTION PLANNING) was developed and established to guide, inform and carry out activities to introduce the business in the market/participants. The communication of the business to participants was structured to communicate:

- the company profile (hardcopy) document
- AS002PT, Application for proficiency testing participation form. It addressed all requirements stipulated in ISO 17043:2023 clause 7.1.2.1 which are the objectives of the PT scheme, criteria to participate, criteria for determining the assigned values and evaluation, etc. It was shared with clients, completed and received back.
- AS001PT, instruction form to participants.

Company website www.azuresky.it.com communicates the business profile including complaints and appeals procedure to be followed, other services, etc.

The business shall communicate any changes in the PT scheme design or operation timely and maintain records to that effect. This is to avoid service impairment and to ensure efficient participation.

3.3 Equipment

To ensure technical consistency and metrological traceability, the Laboratory mandates that all external service providers for equipment and software are selected, approved, and evaluated in accordance with the criteria defined in Checklist 6.6.3.
The process is structured as follows:

- Selection: Potential providers are identified and assessed against the requirements outlined in Checklist 6.6.3.
- Approval: Providers meeting the defined criteria are formally approved and documented.
- Evaluation: Approved providers are subject to periodic re-evaluation using the same checklist to ensure ongoing compliance and suitability.
- Register: All mandated and approved providers are recorded in the Externally Provided Products and Services Register (Section 6.6.3), which serves as the official record of compliance.

This ensures that the Laboratory's reliance on external providers is consistently controlled, traceable, and aligned with established quality management requirements.

3.4. Methods

The laboratory has established coal testing methods covering sampling, proximate analysis, calorific value determination, and sample preparation procedures. Each method explicitly defines and addresses the critical factors required for the correct execution of testing, proficiency testing (PT) and reference material activities, ensuring that potential influences on results are controlled.
Accordingly, all measurement procedure requirements necessary for each testing method are documented in alignment with the following clauses:

- **ISO/IEC 17025**: Clauses 6.4 and 6.5
- **ISO 17034**: Clauses 7.7, 7.8, and 7.9

Clause 7.2.2 ISO 17025: 2017 method validation of the coal testing methods were performed for the following lists of test methods.

- Inherent Moisture analysis
- Volatile matter analysis
- Quick Ash analysis
- Calorific Value analysis
- Total Moisture analysis

A combination of the following techniques were used:

- Calibration or evaluation of bias and precision using reference standards
- Systematic assessment of factors influencing the result
- Testing method robustness through variation of controlled parameters (where necessary)
- Evaluation of measurement uncertainty of the results based on an understanding of the theoretical principles of the method and practical experience of the performance of the test method.

A record of the method validation per test method has a conclusion statement detailing its fitness for the intended use.

NB: In accordance with "**7.2.2.1** The laboratory shall validate non-standard methods, laboratory-developed methods and standard methods used outside their intended scope or otherwise modified. The validation shall be as extensive as is necessary to meet the needs of the given application or field of application". This means that the lab only needs to do validation for lab developed, modified international/national standards. If the standards are used exactly as they are; verification to meet the reproducibility limits within the respective standards should be sufficient.

3.4 Statistical design

For meaningful statistical evaluation, a minimum sample size of n = 10 is required.

Descriptive statistics is to be performed.
ASMF 7.2.2 Statistical Design procedure was developed and implemented.

Statistical treatment

- **Collection of results**: All participants submitted results on an air-dry basis. These results we subsequently converted to a dry basis to ensure comparability across laboratories.
- **Data screening and outlier assessment**: The dataset was examined for consistency and reliability. Outliers were identified using the criterion of ±3 standard deviations from the mean. Results flagged as outliers were excluded from further statistical evaluation. This step ensured that only representative data contributed to the assigned values
- **Statistical evaluation**: For each parameter, the following descriptive statistics were calculated. Mean, standard deviation, RSD, Median, Upper and lower control limits, UoM, standard error, min and max.
- Z-scores were also computed to assess individual laboratory performance relative to the assigned values. Graphical summaries, including population distributions and laboratory-specific plots, were generated to illustrate the spread and comparability of results.
- The procedures to be used to treat participant results from different measurements or test methods which are not technically equivalent, shall not be permitted by the scheme
- **Assignment of reference values**: The CRM certificate values were assigned from the **consensus means** of the participant data set, following removal of statistical outliers. This approach ensures that the reference values are derived from interlaboratory consensus, consistent with international PT practices.
- **Estimation of Measurement Uncertainty**:
  The uncertainty of the assigned reference values incorporates contributions from:
  - Homogeneity assessment
  - Stability Check
  - Characterization Uncertainty
  - Transport effects were evaluated and found to be negligible; therefore, no additional uncertainty component was attributed to transportation.

  Compliance with SANAS R80: This process of reference value assignment ensures that the values are statistically valid, traceable, and supported by an uncertainty budget. It also aligns with the SANAS R80 requirement to clearly document the source of the assigned values and the methods used in their determination.

3.5 Homogeneity and stability assessment of PT items

3.5.1 The assessment of homogeneity and stability shall be performed for every PT round after the PT items have been packaged in their final form. The calculated results shall be included in the PT report.

3.5.2 Criteria for suitable homogeneity is established as follows:

- 10 random samples are extracted from the batch during packaging. These are used for homogeneity assessment prior to the distribution of samples to participants.
- Ash% parameter Test method is used for both purpose of homogeneity test and stability test. Reason is that Ash% parameter in coal is the most studied and stable parameter to be used for homogeneity and stability checks.
- Homogeneity is an important requirement for all PT items and RM items and includes both within- and between-unit homogeneity. The statistics is calculated using ANOVA and it is required to achieve between-unit less than (<) the check value. It ensures that each PT item and each RM item carry the same value for each property.
- Within-unit homogeneity calculated result compared to check value is important to demonstrate that where subsample measurement is assessed against an expected certified reference value (check value). A large difference between the calculated and check value indicates a potential error in the process, the measurement or the formula used. Therefore, a decision can be made to accept the result or stop the process, recalibrate or reject the results.
- Thus this safeguards the risks that inhomogeneity and instability can impact the evaluation of the performance of participants

3.5.3 The assessment for stability shall be performed for the PT items to demonstrated sufficient stability and ensure that they will not undergo any significant change throughout the conduct of the PT round, including storage and transport. A sufficiently stable stability assessment provides confidence to the end user of the RM item to rely on the assigned value at any point within the period of validity of the certificate.

- 10 samples which were used for the homogeneity assessment are analysed to check for stability, the Ash% test method is used for this purpose for the same reason provided previously.
- The mean difference between homogeneity and stability assessments is compared. The difference shall be less than (<) the check value. Then, the PT samples meet stability requirements

3.5.4 A continuous study is performed monthly at the laboratory to determine and conclude stability of RM items under long term storage condition which is (1) one year of opening the bottle. The result will be updated accordingly at the end of the study for all RM units produced.

3.5.5 PT items from previous PT rounds, retained for another PT round, the property values or characteristics to be determined in the PT scheme shall be confirmed prior to distribution.

3.6 Instructions for participants

Distribution of PT items shall be conducted by no later than 10th of every month. Instructions to participants shall be sent together with distribution of PT items no later 10th of every month. Submission of results shall be no later than 25th of every month, however grace period is allowed until 28th of every month until 11pm and participants who couldn't submit on the 25th of every month shall be contacted to submit results.
Feedback report to participants shall be sent no later than 06th of every month

3.6.1 Document AS001PT proficiency testing instruction form and Microsoft link form provides detailed information on the handling and participation in the PT round. The instructions shall include:

- the necessity to treat PT items in the same manner as routine samples, for an example which means instruction to condition PT item for the general analysis for a defined period shall be included.
- ensure to include conditions specified in the respective test method of interest i.e. ISO or SANS or Eskom method or ASTM, etc and meet factors which can influence the measurements or tests of the PT items.
- specific and detailed instructions on the manner of recording and reporting results
- This instruction usually includes parameters such as the units of measurement e.g. "air dry base"
- instructions on return or forwarding of the PT items, a link shall be provided to participants every month
- information on the contact details of the PT provider for enquiries.

3.7 Determination of assigned values

ASMF 7.12 Characterisation shall be used for determining the assigned values for the properties or characteristics of a PT/RM item. Each parameter will have(uncertainty of measurement) that has a guard band stated as follows: The guard band for acceptability of uncertainties is that the uncertainties assigned must be less than the reproducibility limit as per the parameter.

The CRM certificate values shall be assigned from the **consensus means** of the participant data set, following removal of statistical outliers or by using algorithm A (without removing outliers which gives the robust mean). This approach ensures that the reference values are derived from interlaboratory consensus, consistent with international PT practices. An estimate of the uncertainty of the assigned value incorporates contributions from:

- Homogeneity assessment
- Stability Check
- Characterisation Uncertainty

- Transport effects were evaluated and found to be negligible; therefore, no additional uncertainty component was attributed to transportation.

3.8 Disclosure of assigned values

In line with the requirements of ISO 17043:2023 clause 7.2.3.5, measures are in place to ensure that participants cannot gain an advantage through early disclosure of assigned values. The following practices are applied:

- Single submission: each participant is allowed to submit their results once. No edits or re-submissions are allowed.
- Post-closure analysis: All statistical evaluation and assignment of values are performed only after the submission deadline
- Simultaneous release: final PT reports, including the assigned values are issued to all participants at the same time. This ensures that no participant has access to assigned values ahead of others.

The integrity of proficiency testing scheme is maintained through these practices and participants are assured of fair and unbiased evaluation.

3.9 Evaluation and reporting of PT scheme results

Results received from participants shall be recorded and analysed using descriptive statistics. Procedures shall be established and implemented to check the validity of data entry, data transfer, statistical analysis, and reporting.
Microsoft forms links are used to collect and deposit data entry from participants; it then transfers input data into excel workbook. Data analysis shall generate summary statistics and performance statistics and associated information consistent with the statistical design.

The influence of outliers on summary statistics shall be minimized by using the criterion of ±3 standard deviations from the mean. Results flagged as outliers were excluded from further statistical evaluation. The test methods that participants are allowed to participate on are the following:

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Parameter	Result (air-dry basis)	
ISO Ash (ISO 1171)		
Quick Ash		
ISO Volatile (ISO 562)		
ISO Moisture (SANS 5925)		
Calorific Value (ISO 1928)		
Total Sulphur (ASTM D4239)		
Total Moisture (semi-prep: ISO 589)		
Hardgrove Index (ASTM D409)		
Abrasive Index (Eskom method)		
Hydrogen (ISO 5074)		
Total Carbon		
Ash Fusion Temperature	*DT*	
	ST	
	HT	
	FT	

In accordance with SANAS R80, small interlaboratory comparisons are defined as those with seven or fewer participants. As this scheme is not classified as a small ILC, Azuresky applies an internal threshold of n ≥ 10 for the calculation and reporting of z-scores. This criterion is applied to ensure that statistical estimates (mean, standard deviation, and z-scores) are robust and not unduly influenced by small datasets. Where the number of participants is fewer than 10, only indicative results on an air-dry basis are provided.

The PT scheme preparation is structured in a way that no PT item goes to the participants prior to its homogeneity assessment being conducted and found to be sufficient. Stability checks for each PT round robin is strategically placed on the 25th of each month, a date prior to the statistical analysis of PT rounds to ensure that a stability of a sample is conducted and if any instability is identified, the PT round can be rendered null and void. This method provides a criteria and procedure to ensure that the PT items distributed and the collected data are suitable for performance evaluation.

3.10 Evaluation of performance

Interpretation of trends requires a minimum of five data points to ensure a meaningful analysis, while graphical representation necessitates at least two data points to establish a visual comparison.

When the sample size is sufficiently large, the sampling distribution of the sample mean approaches a normal distribution, regardless of the shape of the original population distribution. This assumption underpins all statistical analyses and calculations for the evaluation of performance.

The preferred method of evaluation which meet the objectives of the PT scheme. It is described as follows:
The criteria for evaluation of performance is based on consensus statistics from the current round or previous round, then a robust estimate of the standard deviation of participant results is the preferred statistic. This approach enables the performance score of z-scores and to set the standard deviation for proficiency assessment to the calculated estimate of the standard deviation.

Expert commentary shall be provided on the performance of participants with regard to the following:
a) overall performance against prior expectations, that the z-scores fall within the normal distribution curve
b) variation within and between participants, and comparisons with any previous PT rounds, or published data;
c) variation between measurement or test methods;
d) possible sources of error (with reference to outliers or poor performance) and suggestions for improving performance;
e) advice and feedback to participants as part of the continuous improvement procedures of participants;
f) situations where unusual factors make evaluation of results and commentary on performance impossible;
g) any other suggestions, recommendations or general comments;
h) conclusions.

3.11 Reports

3.11.1 PT reports

PT report template AS-PT001R general analysis was developed and implemented, AS-PT002R for Hardgrove index was developed and implemented. Reports shall be clear, accurate, objective and comprehensive and include data covering the results of all participants, together with an indication of the performance of individual participants**.**

Reports shall include the following:
- the name and contact details (email address);
- identification of person(s) authorizing the report;
- an indication of which activities are provided by external providers when they affect the production or characterization of the PT items or the services provided;

- the date of issue and status (e.g. preliminary, interim, or final) of the report;
- unique identification that all its components are recognized as a portion of a complete report and a clear identification of the end;
- a statement of the extent to which results are confidential;
- a unique identification of the report and the PT scheme;
- a clear description of the PT items used, including necessary details of the PT item's production and homogeneity and stability assessment;
- the results of participants, including the reported measurement uncertainties (if required);
- procedures used to statistically analyse the data;
- statistical data and summaries, including assigned values, range of acceptable results and graphical displays;
- details of the metrological traceability, and uncertainty of any assigned value;
- procedures used to establish any assigned value and its uncertainty;
- assigned values, their uncertainties and summary statistics for measurement or test methods used by each group of participants (if different measurement or test methods are used by different groups of participants);
- procedures used to establish the standard deviation for proficiency assessment, or other criteria for evaluation;
- comments on the performance of participants;
- information about the design and implementation of the PT scheme;
- advice on the interpretation of the statistical analysis;
- comments or recommendations based on the outcomes of the PT round.

ASMF 7.4.3. Participant use of Proficiency Testing report policy was developed and implemented. The laboratory‘s participation in Azuresky PT scheme will be evaluated against their plan. Failure of laboratories to show effective participation could result in suspension of accreditation. The laboratory shall make available to the assessment team all proficiency testing scheme reports.

When it is necessary to issue a new or amended report for a PT round, this report shall include the following:
a) a unique identification;
b) a reference to the original report that it replaces or amends;
c) identification of the amendment and a statement concerning the reason for the amendment or re-issue.
It is important to ensure that when issuing an amended report to a subset of participants, an analysis of the potential impact on the other participants for that PT round shall be made to ensure there is no influence on the general performance of the other participants.

If there is a need to issue a statement of participation or performance in addition to the PT report, the statement shall not be misleading.

3.11.2 Testing reports

The reporting of results will be done in accordance with ISO 17025 clause 7.8, and shall encompass the following:

7.8.1 General Requirements for Reporting Results
7.8.1.1 Results must be reviewed and formally approved before being released.
7.8.1.2 Reports shall present results in a manner that is precise, clear, objective, and free from ambiguity. Reports may take the form of test reports, or sampling reports, and must contain all information agreed upon with the customer, as well as any details required for proper interpretation and by the method applied. All issued reports must be retained as part of the laboratory's technical records.
NB: Reports may be delivered either in print or electronically, provided all requirements are satisfied.
7.8.1.3 Where agreed with the customer, results may be presented in a simplified format. Any information required under clauses 7.8.2 to 7.8.7 that is not included in the customer-facing report must remain readily accessible.
7.8.2 Common Requirements for Reports
7.8.2.1 Unless valid reasons exist for omission, each report shall include at least the following to avoid misunderstanding or misuse:

- Report title (e.g., "Test Report","Sampling Report")
- Laboratory name and address
- Location where the activity was performed (including customer sites or mobile facilities)
- Unique identification ensuring all parts are recognized as one complete report, with clear indication of its conclusion
- Customer name and contact details
- Method used
- Clear description and identification of the item, including condition if relevant
- Date of receipt of item(s) and sampling date, where critical to validity
- Date(s) of laboratory activity
- Date of issue
- Reference to sampling plan/method, if applicable
- Statement confirming results apply only to the tested, calibrated, or sampled items
- Results, with appropriate units of measurement
- Any deviations, additions, or exclusions from the method
- Identification of the person(s) authorizing the report
- Clear indication of results obtained from external providers

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Note: A statement such as “This report shall not be reproduced except in full without laboratory approval” helps prevent misuse.

7.8.2.2 The laboratory is accountable for all information in the report, except where supplied by the customer. Customer-provided data must be clearly identified, and a disclaimer included if such data could affect validity. If the laboratory did not perform sampling, the report must state that results apply only to the sample as received.

7.8.3 Specific Requirements for Test Reports

7.8.3.1 In addition to the requirements in 7.8.2, test reports shall include, where necessary for interpretation:

- Details of specific test conditions (e.g., environmental conditions)
- Statement of conformity with requirements/specifications, where relevant (see 7.8.6)
- Measurement uncertainty, expressed in the same unit or relative terms (e.g., %) when:
    - Relevant to validity or application of results
    - Required by customer instruction
    - Affecting conformity to specification limits
- Opinions and interpretations, where appropriate (see 7.8.7)
- Any additional information required by methods, authorities, or customers

7.8.3.2 If the laboratory is responsible for sampling, test reports shall also meet the requirements of 7.8.5, where necessary for proper interpretation of results.

- Amendments will be done in accordance with clause 7.8.8.
- Reporting of opinions and interpretations fall outside the scope of accreditation for this lab.
- Sampling reports will be done in accordance with 7.8.5.

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4. Management System Requirements

4.1 Control of management system documents

ISO 17043:2023 clause 8.3; ISO 17025:2017 clause 8.3 and ISO 17034 clause 8.4: Document control is a requirement for all companies and must be in place to manage document creation, review, approval, distribution, and archiving.
This helps ensure that the documentation is controlled, consistently maintained, and organized.
A formal document review and approval process shall be utilized to ensure accuracy and compliance with regulatory requirements.

The process flow illustrated on diagram below may be followed by the team members. Where three persons may contribute to the development of the procedure/document or two persons involved (where one person should review and approve the procedure/document).

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4.1.1 Implement Version Control – see the guideline memo doc 8.3.

Version control has been implemented to track and manage document revisions. Procedures have revision table which has ensured to indicate the document version number, the content of the change, who modified the document, the reason for change and date to avoid confusion. The latest version shall be readily identifiable and accessible. The old document shall be retired (file in obsolete folder) to avoid the use of

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outdated documents. Communication of changes to relevant persons shall be conducted to ensure adequate implementation. Evidence of communication shall be retained.

It is imperative that comprehensive training plans must be conducted to educate employees on the purpose, content, and use of documentation. A skill matrix have been developed to assess employees' competencies, identify skill gaps, and support the creation of training plans.
By providing effective training, the organisation equip employees with the necessary skills and knowledge to create, manage, and utilize documentation effectively throughout their roles.

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4.1.2 Ensure Document Accessibility

ISO 17025 clause 8.2.5 and ISO 17043 clause 8.2.3 states that all personnel involved in laboratory/PT activities shall have access to the parts of the management system documentation and related information that are applicable to their responsibilities.
Only authorized personnel (Azuresky team members) have access to the required documentation. The storage is managed through QMWARELIMS12 in Azuresky system folder.
See doc ASMF 7.11 for more information.

4.2 Control of data and information management

- Records are to be archived and made available for audits (10 years).
- All relevant personnel as per job description shall have access to the data and information needed to perform their activities.
- **QMWARE version 12** information management system is used for the collection, processing, recording, reporting, storage or retrieval of data. It shall be validated for functionality, including the proper functioning of interfaces before introduction. Whenever there are any changes, e.g. software configuration, etc, they shall be authorized, documented and validated before implementation.
- **QMWARE version 12** information management system shall:
  a) be protected from unauthorized access;
  b) be safeguarded against tampering and loss;
  c) be operated in an environment that complies with the system supplier, input of data into LIMS is not auto transfer from measuring equipment. Every mass input is captured in LIMS and the operator involved as well. Changes to any mass inputs and results are also captured automatically in LIMS. Such changes are password protected and privileges are only given to management to make such changes in the case an error that may arise during analysis. If any change occurs in a spreadsheet, LIMS automatically flags this spreadsheet each time it is opened.
  d)be maintained in a manner that ensures the integrity of the data and information;
  e) include recording of system failures and the appropriate immediate and corrective actions.
- **QMWARE version 12** information management system shall ensure tutorial videos which serve as instructions, manuals and reference data relevant to the laboratory information management system are made readily available to personnel.
- Calculations and data transfers shall be checked.

4.3 Actions to address risks and opportunities

A risk assessment to identify risks and opportunities associated with the testing, PT and RM production is to be undertaken annually by the relevant personnel involved in the running of the laboratory.
When there are plans to introduce significant changes to activities which can affect the technical operations of the laboratory, risk assessment shall be performed to identify and manage the risk to ensure that the quality of work is maintained.
See documents ASRI 4.2 and ASRI 8.5-001.

4.4 Improvement

It is in the best interest of Azuresky (Pty) Ltd to improve its processes and running of the laboratory. Amongst other benefits is to seek increased efficiency, improved customer satisfaction, and better regulatory compliance. Therefore, the guideline of ISO 17043:2023 clause 8.6; ISO 17025:2017 clause 8.6 and ISO 17034:2016 clause 8.8 shall be performed.
Opportunities arising from but not limited to the following list shall be performed.

- Management review meeting which is to be held annually.
- Company objectives
- Audit results
- Suggestions from personnel
- Proficiency testing results
- Review of instructions to work/operational procedures
- Corrective actions
- Risk assessment
- Analysis of data
- External assessment e.g. Customer feedback form AS006MF was developed and implemented for use to engage with clients and gather feedback of the service.

Actions shall be planned, implemented and monitored.

Customer Feedback

Feedback from external assessment shall be analysed and used to improve the management system, testing, PT, and RM activities and customer service using **FB01-Evaluation of customer feedback**. This will include consolidating the category scores in the quantitative analysis section to obtain the laboratory averages for each category, then adding the summaries of each customer's comments in the qualitative analysis table in section 2. If any category under the quantitative analysis section gets a score of 2 or below, it will require the lab to assess the risk of the said category using the 8.5 risks and opportunities spreadsheet and then make use of section 3 to identify the area needing improvement, the action to be taken after risk analysis, the responsible

personnel and the target date of the action. A score of 3 will require management to visually & verbally monitor the affected area and a score of 4-5 will warrant a successful period for the lab.

4.5 Corrective Actions

Document F03-09 was adopted and implemented to record the non-conformities that may arise during laboratory activities(operational & managerial). If necessary, risks and opportunities will be identified, assessed, implemented and updated.
Corrective actions shall form item of discussion during the management review meeting and establish improvement and effectiveness of the corrective actions implemented.
For detailed process flow see ASMfF7.17 NC work procedure

4.6 Internal audits

For the internal audits plan see document AUI001 . It shall be updated as necessary in terms of scheduling.
The criteria for an independent person to conduct an audit, a competent experienced person in the field will be consulted to conduct the internal audit however this will be dependent on the availability of funds. Operations Manager will conduct the audit in case of financial constraint.

When audit findings cast doubt on the effectiveness of the operations, Azuresky Management shall timely correct the deviations and shall notify in writing the customers whose activities may have been adversely affected.

4.7 Management reviews

Management review meetings shall be conducted annually. The inputs as described in ISO 17043:2023 clause 8.9.2; ISO 17034:2016 clause 8.6.1 and ISO 17025:2017 clause 8.9.2 shall be discussed.

The purpose of management review is to ensure continuing suitability and effectiveness and introduce necessary changes and improvements.

A record of the meeting minutes shall be prepared, reviewed and signed by the Operations Manager and/or the Laboratory Manager. This is to confirm the contents of the minutes as true facts of discussion.

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Revision

Revision no	Content changed	Reason for revision	Date changed
01	The manual has been rephrased to include testing services offered	The external audit revealed a gap within the system that needed to be addressed	15/04/2026
01	Include annual completion of declaration of interest document no. ASMF 4.2.1 by all personnel	Effective monitoring of the requirement	20/04/2026
01	Added a detailed section that explains how the evaluation of customer feedback is done, under 4.4.	To fully address customer feedback, Clause 8.6.2 of ISO 17025.	22/04/2026
01	Consolidated externally provided products and services (2.3) to align with equipment procedure 6.4 & checklist 6.6.3	Alignment with the revised management system	22/05/2026
01	Removed table under 3.3	Repetitive information on what was already discussed in other sections of the doc.	28/05/2026

Disclaimer: The rev status for this document has been kept at 02 throughout the update of certain sections; this is to avoid discrepancies in evidence submitted for NCs in all areas where this quality manual is concerned. This only applies to the duration of closing SANAS NCs raised during the initial assessment.

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